

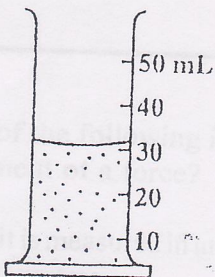
TIME: 1 ¼ HOURS

ANSWER ALL SIXTY (60) QUESTIONS

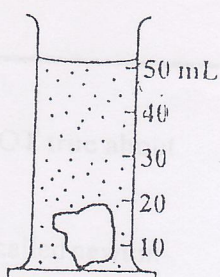
(Only the letter A, B, C or D corresponding to the answer of your choice is required)

- 2 -

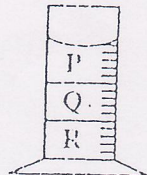
1. The unit of electrical resistance may be expressed as
 - (A) $1 \Omega = 1 \text{ V A}^{-1}$
 - (B) $1 \Omega = 1 \text{ A V}$
 - (C) $1 \Omega = 1 \text{ A V}^{-1}$
 - (D) $1 \Omega = 1 \text{ W A}^{-1}$
2. Which of the following is NOT a fundamental quantity?
 - (A) Temperature
 - (B) Heat energy
 - (C) Length
 - (D) Time
3.



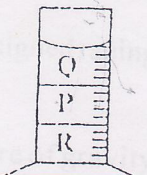
100 g
Figure 1



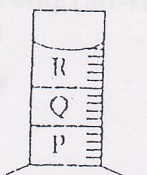
160 g
Figure 2
4. Which of the following is suitable for measuring the diameter of a human hair?
 - (A) Tape measure
 - (B) Vernier calipers
 - (C) Micrometer screw gauge
 - (D) Metre rule
5. Three immiscible liquids P, Q and R have densities which differ. Q is denser than P, less dense than R. Which of the following diagrams shows how the liquids settle in a measuring cylinder?



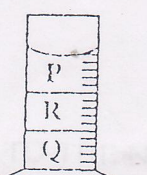
(A)



(B)



(C)

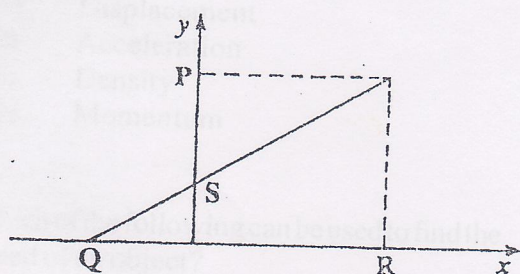


(D)

Figure 1 shows a measuring cylinder containing a fixed volume of liquid. Figure 2 shows a solid body submerged in the liquid. In each case, the total mass of the container and its contents is indicated. The density of the solid is

- (A) 0.30 g mL^{-1}
- (B) 0.33 g mL^{-1}
- (C) 3.00 g mL^{-1}
- (D) 3.20 g mL^{-1}

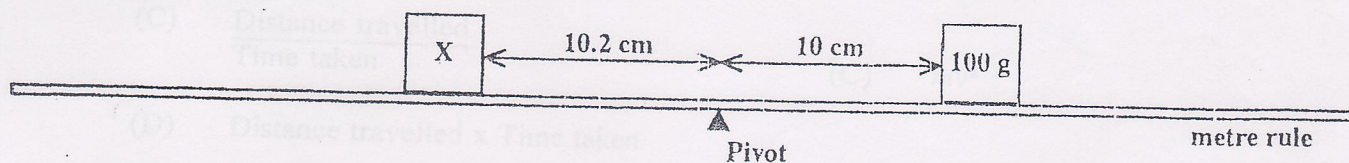
Item 6 refers to the diagram below. 6.



When $x = 0$, the value of y is

- (A) Q
- (B) S
- (C) P
- (D) P/R

7.



The diagram above represents a 100 g mass which can be balanced by placing a mass X at the position shown. If Mass X is to be used to balance the 100 g, it should be

- (A) less than 100 g
- (B) equal to 100 g
- (C) a little greater than 100 g
- (D) about 200 g

8. Which of the following is NOT true about the moment of a force?

- (A) It is measured in units called newton-metres.
- (B) It is the amount of force needed to keep a body turning.
- (C) It is the turning effect that a force has when it acts on a body.
- (D) It is the product of the force and the perpendicular distance of its line of action from a point.

9. A stable well-designed racing car must have a

- (A) low centre of gravity
- (B) very large mass
- (C) very high drag (air-resistance)
- (D) long front

10. Which of the following is NOT a vector quantity?
- (A) Displacement
 - (B) Acceleration
 - (C) Density
 - (D) Momentum
11. Which of the following can be used to find the speed of an object?
- (A) $\frac{\text{Change in velocity}}{\text{Time taken}}$
 - (B) $\frac{\text{Change in displacement}}{\text{Time taken}}$
 - (C) $\frac{\text{Distance travelled}}{\text{Time taken}}$
 - (D) Distance travelled x Time taken
12. When two bodies collide, momentum is conserved. This means that the
- (A) kinetic energy before impact is equal to that after impact
 - (B) momentum of each body is unchanged after impact
 - (C) algebraic sum of the velocities before impact is equal to the sum of the velocities after impact
 - (D) total momentum of the bodies before impact is equal to the total momentum of the bodies after impact
13. A book of weight 12 N is resting on a table. The force that the table exerts on the book is
- (A) 0 N
 - (B) greater than 0 N but less than 12 N
 - (C) 12 N
 - (D) greater than 12 N
14. An object is removed from the ground and placed on a shelf. Which of its properties would you expect to change?
- (A) Mass
 - (B) Volume
 - (C) Potential energy
 - (D) Kinetic energy
15. The kinetic energy of a body of mass, m , and velocity, v , is
- (A) $\frac{m}{v}$
 - (B) mv
 - (C) mv^2
 - (D) $\frac{mv^2}{2}$
16. A liquid is contained in a beaker. It is true to say that the pressure
- (A) is greatest at the bottom of the liquid
 - (B) is greatest at the surface of the liquid
 - (C) is the same at all levels in the liquid
 - (D) exerted by the liquid is the same as atmospheric pressure
17. Which of the following units is used to measure the force per unit area on a surface?
- (A) The newton (N)
 - (B) The watt (W)
 - (C) The joule (J)
 - (D) The pascal (Pa)

18. Which of the following is longitudinal in nature?

- (A) Infrared rays
- (B) Light rays
- (C) Sound waves
- (D) Water waves

19. Which of the following is responsible for the glasshouse effect?

- (A) Microwaves
- (B) Ultraviolet rays
- (C) Light rays
- (D) Infrared rays

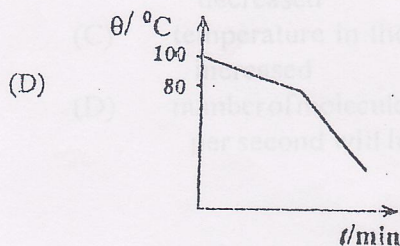
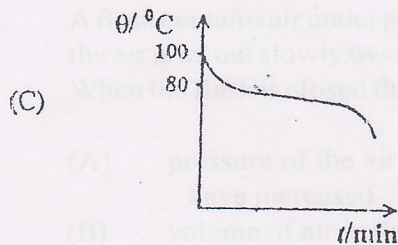
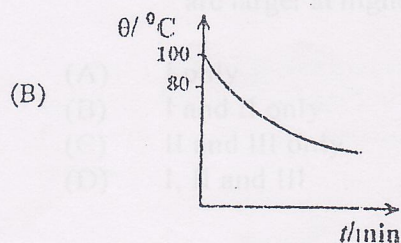
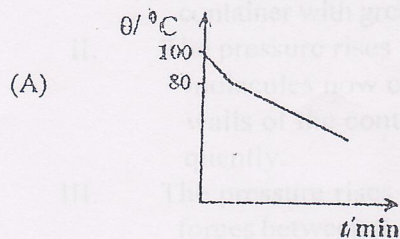
20. Which of the following is correct?

- (A) $T/K = \theta/^{\circ}\text{C} = 273$
- (B) $T/K = \theta/^{\circ}\text{C} + 273$
- (C) $T/K = 273 \theta/^{\circ}\text{C}$
- (D) $273 T/K = \theta/^{\circ}\text{C}$

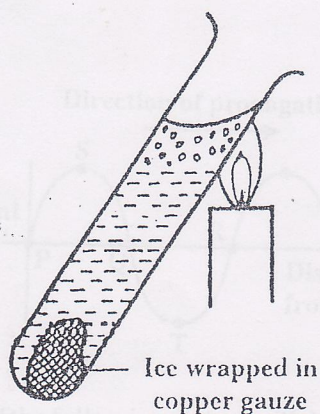
21. The specific heat capacity of a material is the energy required to

- (A) melt 1 kg of the material with no change of temperature
- (B) change the temperature of the material by 1 K
- (C) change 1 kg of the liquid material to 1 kg of gas without a change in temperature
- (D) change the temperature of 1 kg of the material by 1 K

22. Some molten naphthalene at 100°C is allowed to cool down at room temperature. If naphthalene has a melting point of 80°C , which of the following BEST represents the cooling curve?



23.



The figure above shows a piece of ice wrapped in copper gauze and submerged in a glass test tube of water. The water when heated at the top boils long before the ice is melted. The MAIN reason for this is that

- (A) the test tube is a poor conductor of heat
- (B) copper is a good conductor of heat
- (C) ice is less dense than water
- (D) water is a poor conductor of heat

24.

An electric kettle full of water is plugged into the mains. The process by which heat travels through the water is

- (A) electrification
- (B) evaporation
- (C) convection
- (D) radiation

25.

When a gas is heated in a closed container at a constant volume, the pressure of the gas rises. Which of the following statements concerning this fact is/are TRUE?

- I. The pressure rises because the gas molecules strike the walls for the container with greater momentum.
- II. The pressure rises because the gas molecules now collide with the walls of the container more frequently.
- III. The pressure rises because the forces between the gas molecules are larger at higher temperatures.

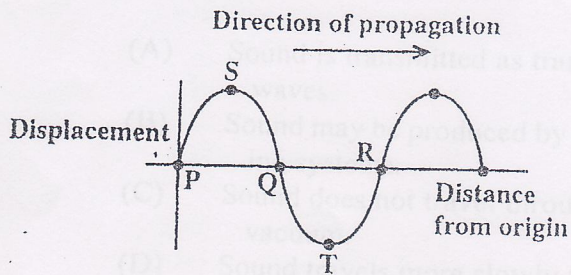
- (A) I only
- (B) I and II only
- (C) II and III only
- (D) I, II and III

26.

A flask contains air under pressure. Some of the air is let out slowly over a period of 10 s. When the flask is closed the

- (A) pressure of the air in the flask will have increased
- (B) volume of air in the flask will have decreased
- (C) temperature in the flask will have increased
- (D) number of molecules striking the wall per second will have decreased

27.

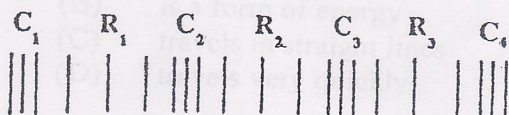


Which of the following statement(s) about the wave shown in the diagram above is/are TRUE?

- I. Points P, Q and R are in phase.
- II. Points S and T are out of phase.
- III. The wavelength of the wave is the distance PR.

- (A) I only
- (B) II only
- (C) I and II only
- (D) II and III only

28.



The diagram above represents a longitudinal wave where, C and R represent compressions and rarefactions respectively. If λ is the wavelength, the distance between C_1 and R_2 is

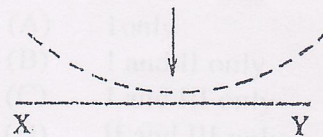
- (A) $\frac{1}{2}\lambda$
- (B) $1\frac{1}{2}\lambda$
- (C) 1λ
- (D) 2λ

29.

Plane water waves which travel from deep water to shallow water

- (A) are diffracted
- (B) increase their speed
- (C) undergo a change in frequency
- (D) undergo a change in wavelength

30.



The figure above shows a wavefront about to reach a reflecting surface XY. Which of the following would represent the wavefront after reflection?

- (A)
- (B)
- (C)
- (D)

31. Which of the following statements about sound is NOT correct?

- (A) Sound is transmitted as transverse waves.
- (B) Sound may be produced by vibrating systems.
- (C) Sound does not travel through a vacuum.
- (D) Sound travels more slowly than light.

32. Which of the following types of radiation is responsible for an atom of $^{14}_6\text{C}$ changing into an atom of $^{14}_7\text{N}$?

- (A) α
- (B) β
- (C) γ
- (D) Infrared

33. The sharp edges of shadows suggest that light

- (A) has a wave nature
- (B) is a form of energy
- (C) travels in straight lines
- (D) travels very quickly

34. The table below lists the refractive indices of four different materials

Material	Refractive Index
Air	1.0
Ice	1.3
Perspex	1.5
Diamond	2.4

In which medium would the light waves have the slowest speed?

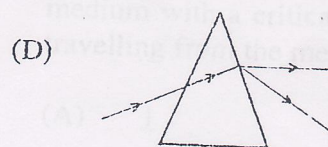
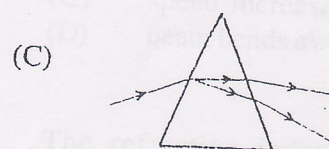
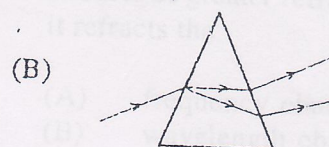
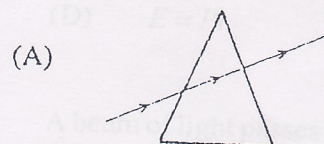
- (A) Air
- (B) Ice
- (C) Perspex
- (D) Diamond

35. Which of the following would be true of the image of an object placed at the bottom of a tank of water and viewed vertically from above?

- I. It is virtual.
- II. It is diminished.
- III. It is nearer to the eye than the object.

- (A) I only
- (B) I and II only
- (C) I and III only
- (D) II and III only

36. Which of the diagrams BEST represents the passage of a beam of white light through a triangular glass prism?



37. The image on the film of a camera is usually

- (A) inverted, diminished, real
- (B) inverted, diminished, virtual
- (C) upright, diminished, real
- (D) upright, magnified, real

38. For which of the following object distances will a convex lens of focal length 18 cm produce a real image?

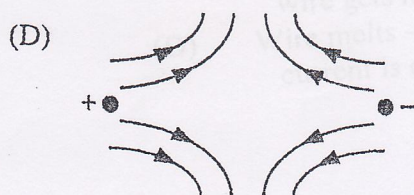
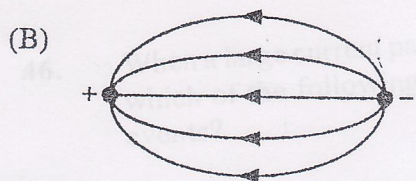
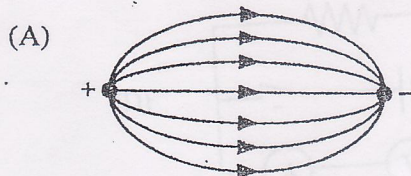
I. 15 cm
II. 36 cm
III. 54 cm

- (A) I only
(B) I and II only
(C) II and III only
(D) I, II and III

39. A plastic rod, P, is rubbed with cloth, Q. P becomes positively charged. This is because

- (A) protons flowed from Q to P
(B) protons flowed from P to Q
(C) electrons flowed from Q to P
(D) electrons flowed from P to Q

40. Which of the following diagrams represents the electric field existing between two oppositely charged point charges?



41. The current in a wire is one ampere if a charge of

- (A) 10 coulombs flows through it for 10 seconds
(B) 1 coulomb flows through it in 10 seconds
(C) 10 coulombs flow through it in 1 second
(D) 100 coulombs flows through it in 10 seconds

42. Which of the following formulae (in which the symbols have their usual meaning) is a mathematical expression of Ohm's Law?

- (A) $V = IR$
(B) $P = VI$
(C) $Q = It$
(D) $E = Pt$

43. A beam of light passes from one medium to another of greater refractive index. When it refracts the

- (A) frequency changes
(B) wavelength changes
(C) speed increases
(D) beam bends away from the normal

44. The refractive index of a transparent medium with a critical angle, c , for light travelling from the medium to air is

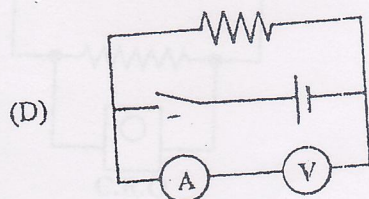
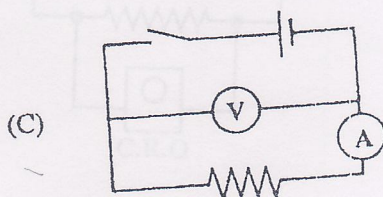
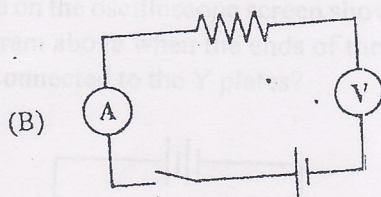
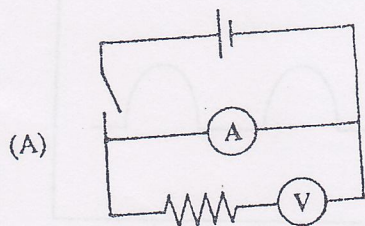
(A) $\frac{1}{c}$

(B) $\frac{90^\circ}{\sin c}$

(C) $\frac{\sin 90^\circ}{\sin c}$

(D) $\sin c$

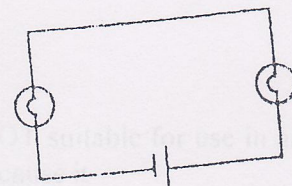
45. A student requires a circuit to measure the resistance of a resistor. Which of the circuits below is correctly connected?



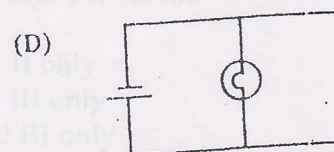
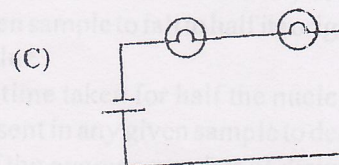
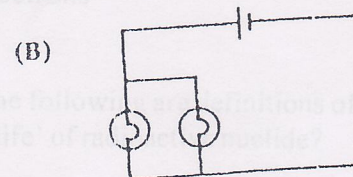
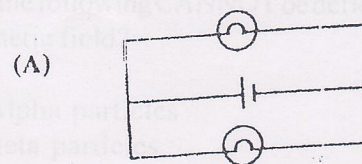
46. When a large current passes through a fuse, which of the following is the sequence of events?

- (A) Wire gets hot → wire melts → current is cut off
 (B) Wire gets hot → current is cut off → wire melts
 (C) Wire melts → current is cut off → wire gets hot
 (D) Wire melts → wire gets hot → current is cut off

47.



The diagram above shows an electric circuit with a cell and two filament bulbs. Which of the following circuits is the same as the circuit above?

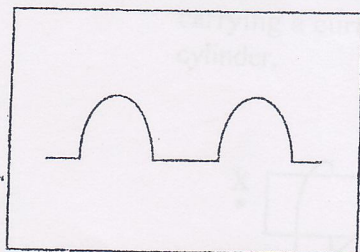


48. Rectification can BEST be done by

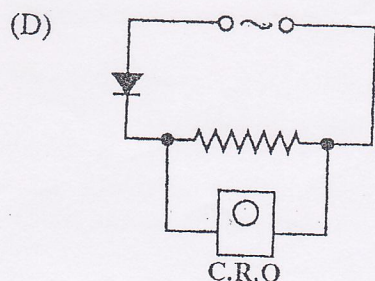
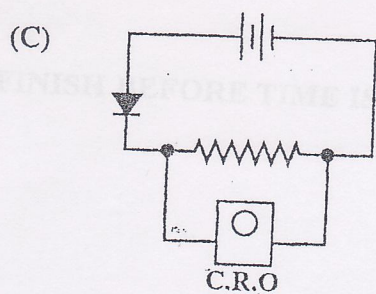
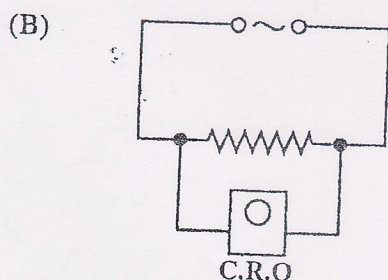
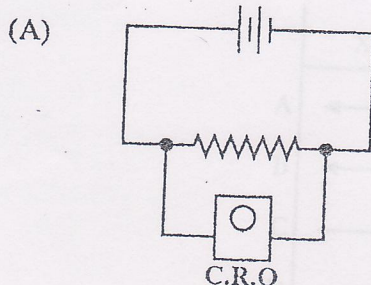
- (A) transformer
 (B) capacitor
 (C) transistor
 (D) diode

GO ON TO THE

49.



Which of the circuits below could produce the trace on the oscilloscope screen shown in the diagram above when the ends of the resistor are connected to the Y plates?



50. Steel is NOT suitable for use in an electro-magnet because it

- (A) is too difficult to magnetise
- (B) retains magnetism too well
- (C) loses its magnetism too easily
- (D) contains too many domains

51. Which of the following CANNOT be deflected by a magnetic field?

- (A) Alpha particles
- (B) Beta particles
- (C) Gamma rays
- (D) Electrons

52. Which of the following are definitions of the term 'half-life' of radioactive nuclide?

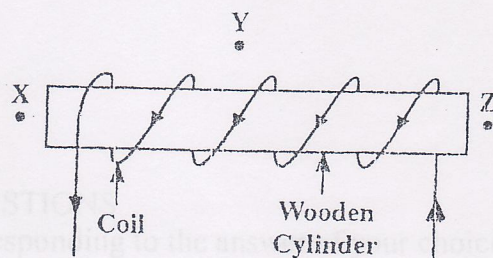
- I. The time taken for the activity of any given sample to fall to half its original value
- II. The time taken for half the nuclei present in any given sample to decay
- III. Half the average number of disintegrations per second

- (A) I and II only
- (B) I and III only
- (C) II and III only
- (D) I, II and III

53. Which of the following persons is associated with nuclear energy?

- (A) Thompson
- (B) Bohr
- (C) Chadwick
- (D) Einstein

Item 60 refers to the following diagram of a coil carrying a current and wrapped around a wooden cylinder.



60. Which row of the table below shows the magnetic field directions at X, Y and Z?

	X	Y	Z
A	←	→	←
B	←	←	→
C	→	←	←
D	→	←	→

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.